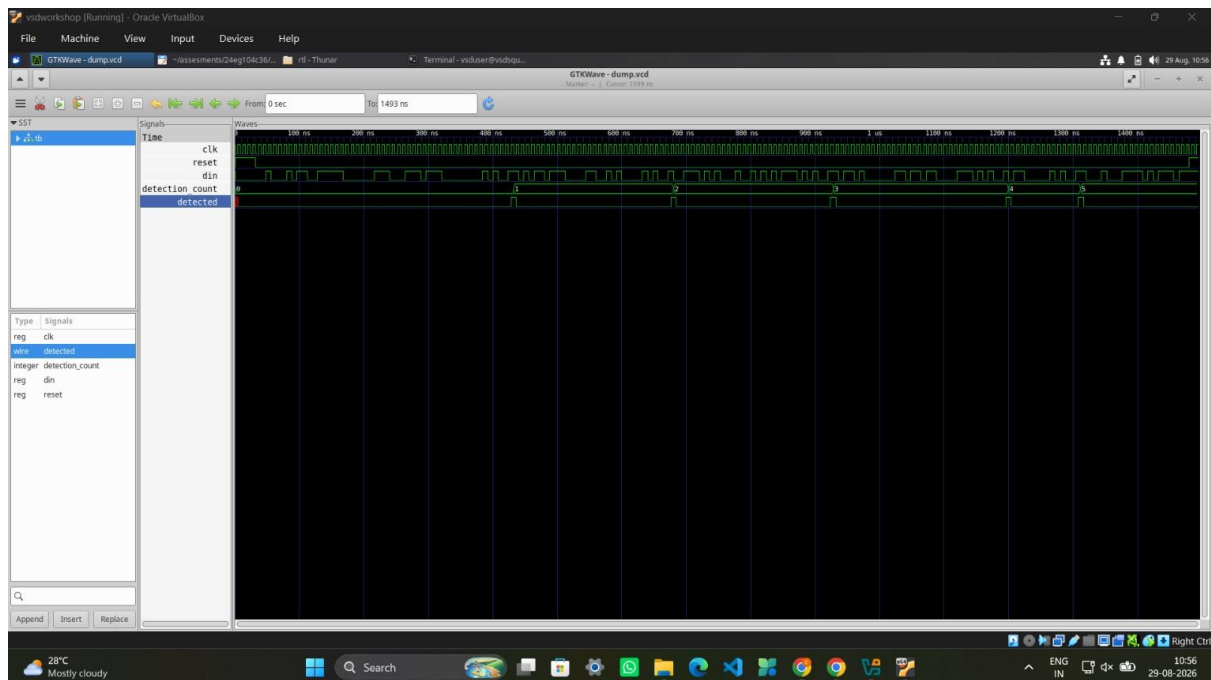


RTL vs Gate-Level Simulation Comparison

The RTL simulation and Gate-Level Simulation (GLS) were compared using the same testbench and input sequence. The target sequence 0101001 was successfully detected in both simulations.

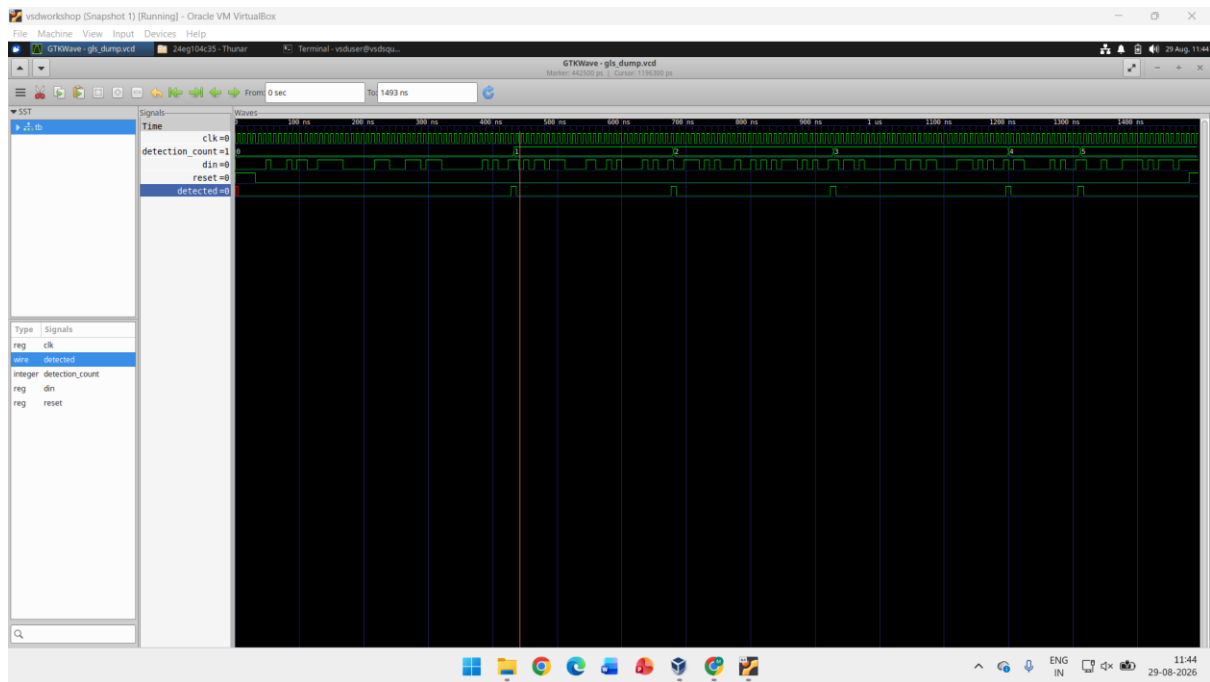
RTL Simulation Result

- First detection occurred at approximately **428 ns**
- Total detection pulses: **5**
- Detection count reached **5**



Gate-Level Simulation Result

- First detection occurred at approximately **429 ns**
- Total detection pulses: **5**
- Detection count reached **5**



Parameter	RTL Simulation	GLS
Target sequence	0101001	0101001
First detection	~428 ns	~429 ns
Total detections	5	5

Final Conclusion

The synthesized implementation preserves the functional behavior of the RTL design for the given testbench. Both RTL and Gate-Level Simulation detect the same input sequence and produce five detection pulses. The small timing difference observed between RTL and GLS does not affect the logical functionality of the sequence detector.